EECE.5494. MODEL-BASED SYSTEMS ENGINEERING

OOSEM, MagicGrid

Ola Batarseh, PhD
Dassault Systèmes

© Dassault Systèmes | Confidential Information | 3/20/2026

			Pillar			
Domain			Requirements	Structure	Behavior	Parameters
	Problem	Black Box	Stakeholder Needs	System Context	Use Cases	Measures of Effectiveness
		White Box		Conceptual Subsystems	Functional Analysis	MoEs for Subsystems
	Solution		System Requirements	System Structure	System Behavior	System Parameters
			Subsystem Requirements	Subsystem Structure	Subsystem Behavior	Subsystem Parameters
			Component Requirements	Component Structure	Component Behavior	Component Parameters
	Implementation		Implementation Requirements			

© Dassault Systèmes | Confidential Information | 3/20/2026

STAKEHOLDER NEEDS

- The fundamental requirements and expectations from stakeholders for the system.
- How stakeholder needs influence all subsequent steps.

- Types of stakeholders: Users, Customers, Regulators, Developers, etc.
- Examples of stakeholder needs (e.g., usability, safety, reliability).

© Dassault Systèmes | Confidential Information | 3/20/2026

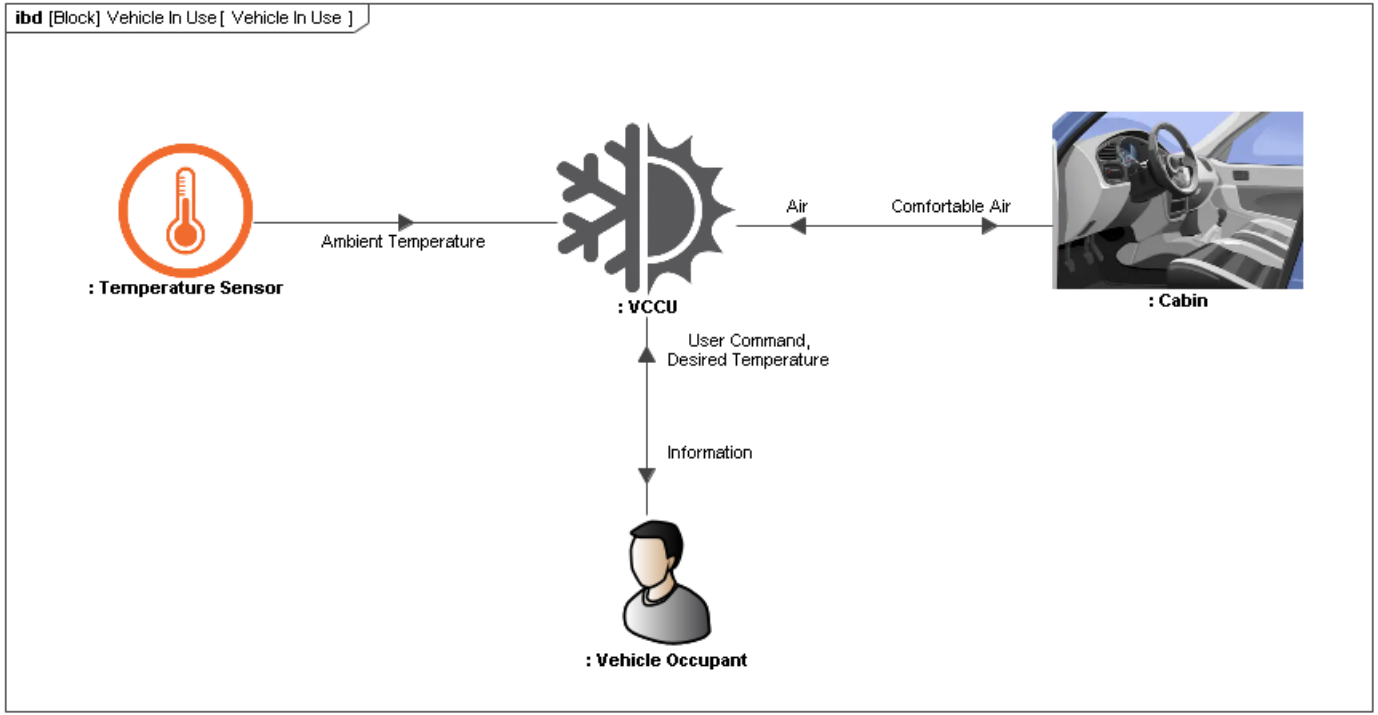
CAPTURE STAKHOLDERS NEEDS AS REQUIREMENTS

#	Name	Text
1	SN-1 Stakeholder Needs	
2	SN-1.1 User Needs	
3	SN-1.1.1 Sound Level	Climate control unit in max mode shall not be louder than engine.
4	SN-1.1.2 Manual Control	I should be able to start and stop climate control by myself.
5	SN-1.1.3 Heating & Cooling	The unit must be able to heat and cool.
6	SN-1.1.4 Energy Consumption	I'd like the unit to consume as little energy as possible.
7	SN-1.1.5 Ambient Temperature	I want to see the ambient temperature on the screen or some other output device.
8	SN-1.1.6 Desired Temperature	It should be a possibility to easily specify the desired temperature.
9	SN-1.1.7 Comfortable Temperature	I'd like to feel comfortable temperature while being in the cabin.
10	SN-1.2 Design Constraints	
11	SN-1.2.1 Total Mass	Mass of the unit shall not exceed 2 percent of the total car mass.

© Dassault Systèmes | Confidential Information | 3/20/2026

BLACK BOX - STRUCTURE

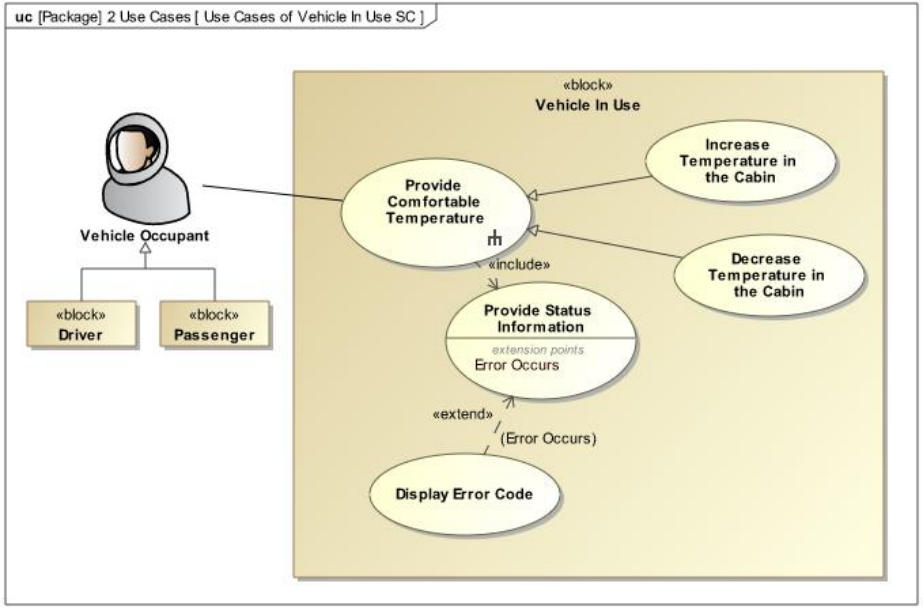
- System context is specified and represented in the SysML internal block diagram (ibd)



© Dassault Systèmes | Confidential Information | 3/20/2026

BLACK BOX– USE CASES

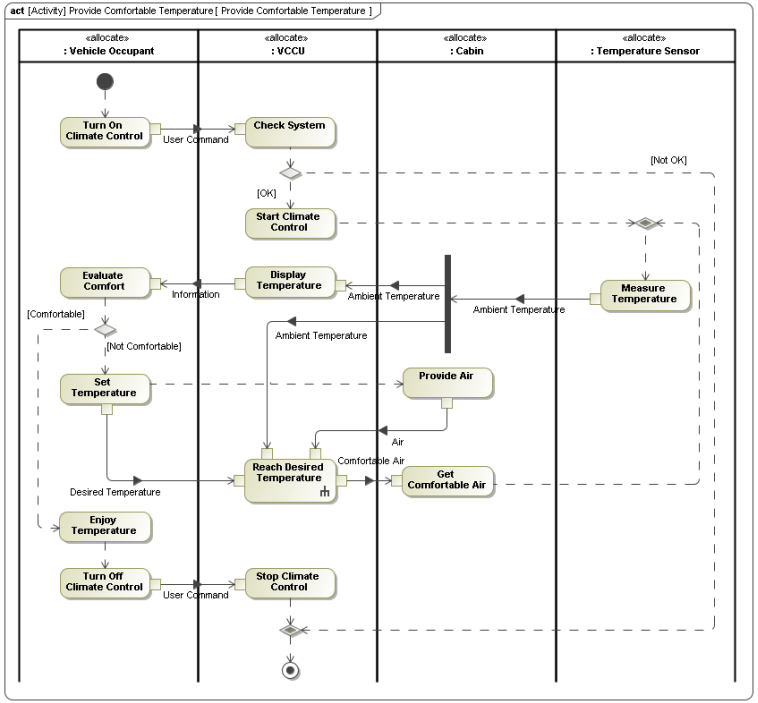
- The System Context provides an external perspective of the system, defining the boundary between the system itself and external elements, such as actors like humans or other systems. For instance, this could include scenarios like a vehicle in operation or a train at a station. In the model, the System Context is depicted as a block, shown as a box with the specific system context's name displayed at the top.



© Dassault Systèmes | Confidential Information | 3/20/2026

BLACK BOX- USE CASE SCENARIO

- The Use Case Scenario (main success scenario) outlines the steps required to execute a use case within a business process.
  - These steps are described in a sequence of sentences, and logical gates can be incorporated to represent alternative flows.
  - The scenario may also involve decisions and branches to account for different execution paths.
- Additionally, Use Case Scenarios can be detailed further with behaviors, such as activities and interactions.



© Dassault Systèmes | Confidential Information | 3/20/2026

BLACK BOX – TRACE BEHAVIORS TO STRUCTURE

Identify and allocate structural elements of the black box to specific actions, defining how each component supports or interacts with the system's operations and objectives.

Legend

↗ Allocate

Provide Comfortable Temperature(classifier	Vehicle In Use	: Cabin	: Temperature Sensor	: WCCU	: Vehicle Occupant
		2	1	5	5
:Check System	1			↗	
:Display Temperature	1			↗	
:Enjoy Temperature	1				↗
:Evaluate Comfort	1				↗
:Get Comfortable Air	1	↗			
:Measure Temperature	1		↗		
:Provide Air	1	↗			
:Reach Desired Temperature	1			↗	
:Set Temperature	1				↗
:Start Climate Control	1			↗	
:Stop Climate Control	1			↗	
:Turn Off Climate Control	1				↗
:Turn On Climate Control	1				↗

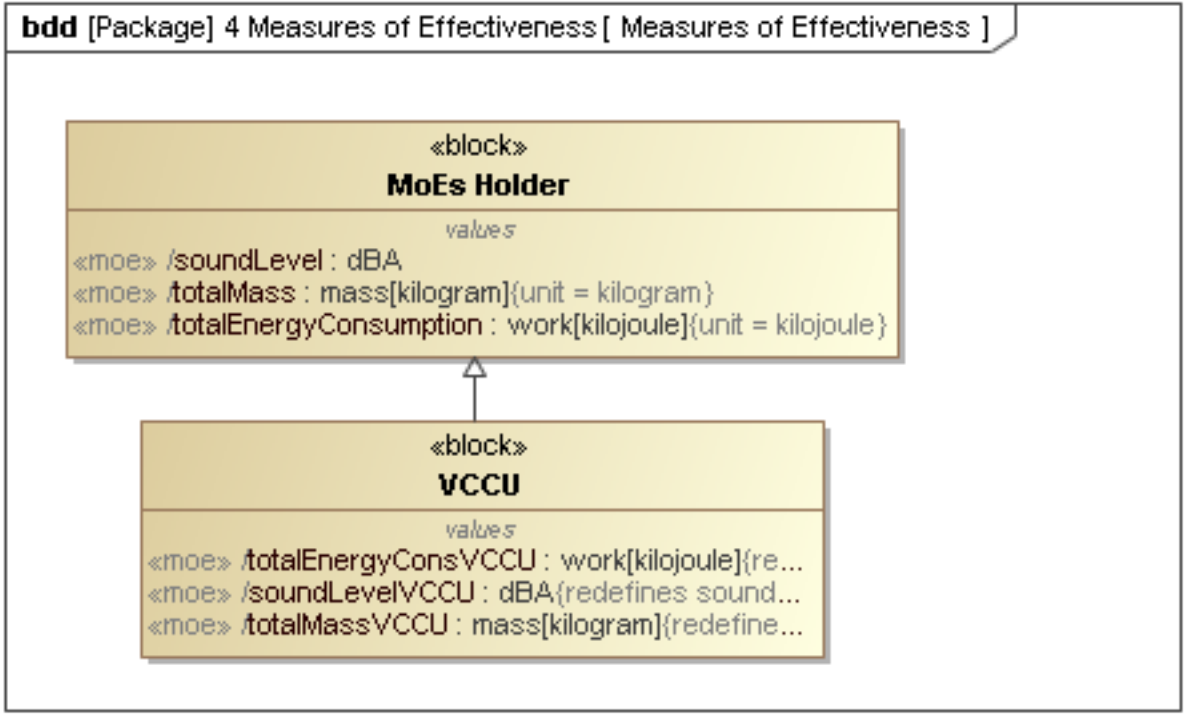
© Dassault Systèmes | Confidential Information | 3/20/2026

UMASS LOWELL

DASSAULT SYSTEMES

BLACK BOX – MEASURE OF EFFECTIVENESS

- Assess how well the black box accurately defines and manages interactions, inputs, outputs, and functional requirements to ensure that it meets stakeholder needs and operational goals



© Dassault Systèmes | Confidential Information | 3/20/2026

IF WE™

© Dassault Systèmes | Confidential Information | 3/20/2026

DASSAULT SYSTEMES

DISCOVER MORE ON 3DS.COM

© Dassault Systèmes | Confidential Information | 3/20/2026

DASSAULT SYSTEMES